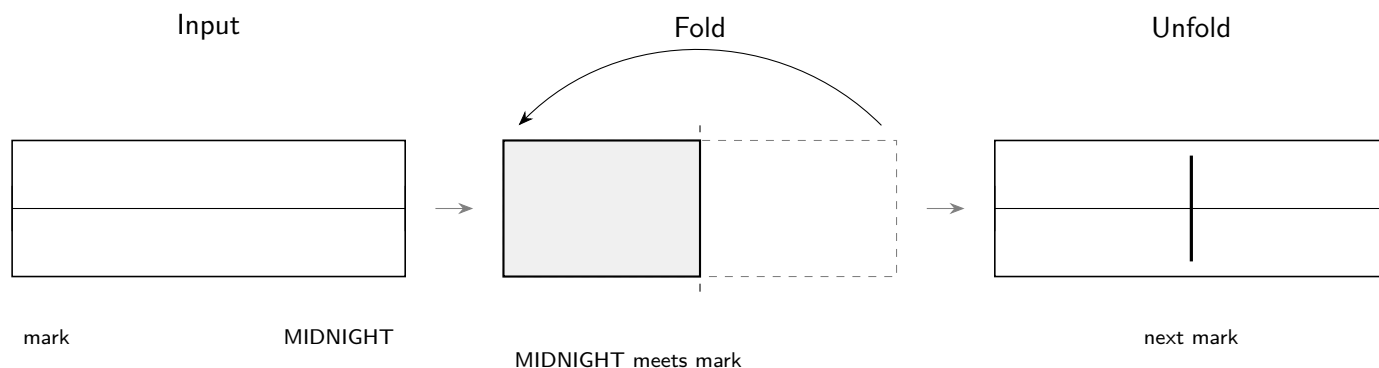
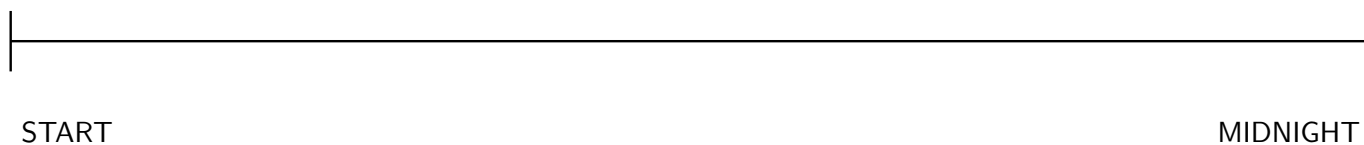


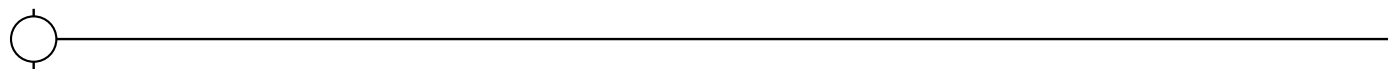
Start with the card OFF. The first switching mark is halfway from START to MIDNIGHT. Each next switching mark is halfway from the previous mark to MIDNIGHT. Flip the card at every switching mark. This rule keeps going before midnight.



Problem 1: Make the first four switching marks on a paper strip. Play the rule with the card. Can the rule have a last switching mark before midnight?



Problem 2: Use the same switching marks. The card starts OFF. A counter starts at START. At each switching mark, flip the card and move the counter halfway from its present place toward END. As the rule keeps going, does the card settle on one side? What place does the counter approach?



START

END

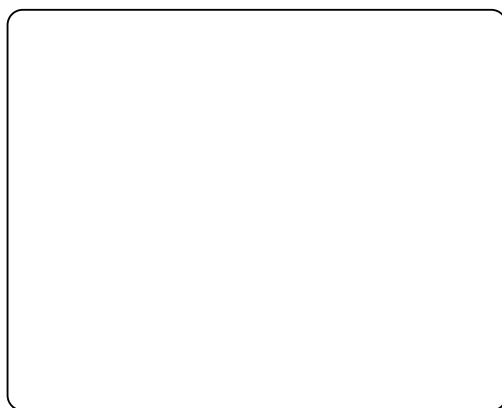
Card

Counter

Problem 3: What, if anything, does the switching rule alone tell you to show at midnight? Use the card and switching marks to support your answer.



MIDNIGHT

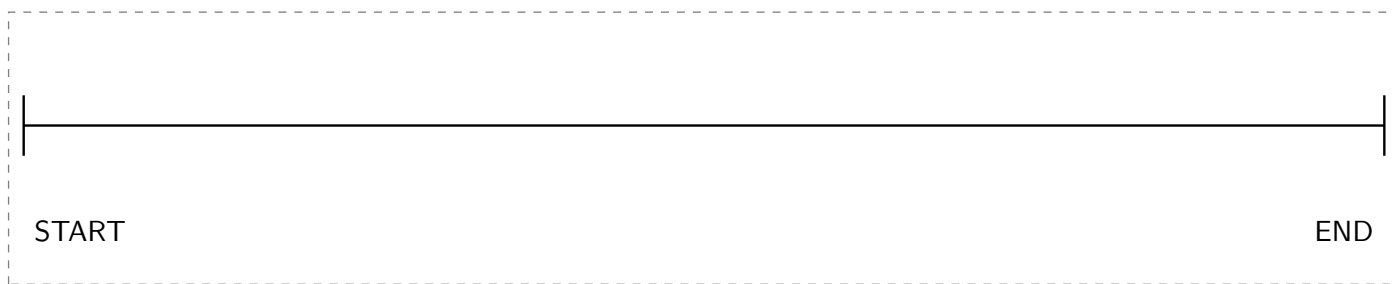


Problem 4: Keep every switch before midnight. Make two different extra rules for the card at midnight. Decide whether either extra rule follows from the original switching rule.

At midnight:

At midnight:

Problem 5: Someone adds this rule: “At midnight, use the state that the earlier states approach.” Try that rule for the ON/OFF card and for the moving counter. Does it assign a state in each case?



ON

OFF